

ROC AUC Study Guide

OUD Relapse Prediction Context

The Setup

We have a logistic regression model predicting whether a patient with OUD will relapse after treatment. The outcome is binary:

1 = relapsed 0 = stayed clean

The model does not output a hard 0/1 — it outputs a **probability** for each patient: e.g., 0.73, 0.21, 0.55. To make a prediction you choose a **threshold**. Default is 0.5: above = predict relapse, below = predict clean.

The 2×2 Confusion Matrix

	Predicted Relapse (1)	Predicted Clean (0)
Actually Relapsed (1)	True Positive (TP)	False Negative (FN)
Actually Clean (0)	False Positive (FP)	True Negative (TN)

Sensitivity and Specificity

Sensitivity — of everyone who actually relapsed, how many did we catch?

$$TP / (TP + FN)$$

Miss a relapser = False Negative. Low sensitivity = lots of missed relapses. In OUD, missing a relapse is costly — that patient doesn't get extra intervention.

Specificity — of everyone who stayed clean, how many did we correctly leave alone?

$$TN / (TN + FP)$$

Wrongly flag a clean patient = False Positive. Low specificity = unnecessary alarms.

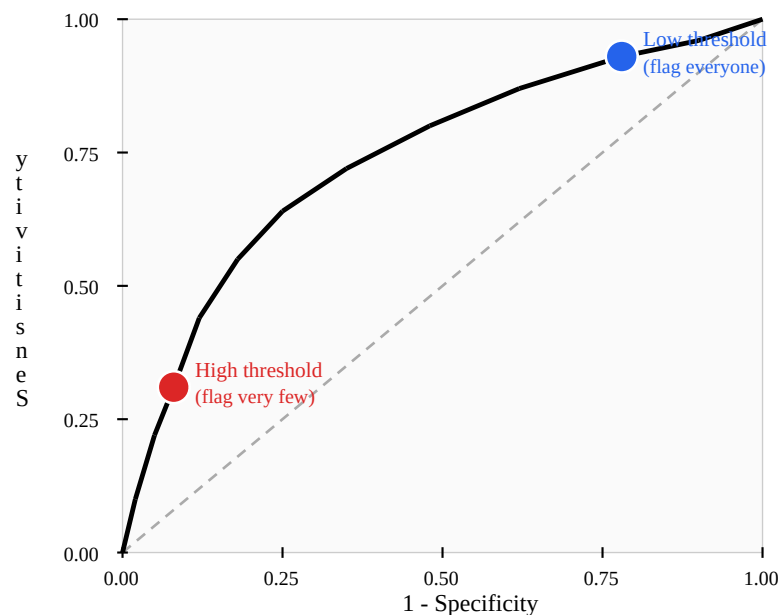
The Threshold Tradeoff

Threshold	Who gets flagged	Sensitivity	Specificity
Low (e.g. 0.3)	Almost everyone	High — catch most relapsers	Low — many false alarms
High (e.g. 0.7)	Very selective	Low — miss many relapsers	High — few false alarms

As threshold increases: sensitivity decreases monotonically, specificity increases monotonically.

The ROC Curve

Instead of picking one threshold, sweep it from 0 to 1 and plot every resulting (sensitivity, 1–specificity) pair. Each point on the curve is one threshold value.



Blue dot = low threshold: flag almost everyone. High sensitivity, low specificity. **Red dot** = high threshold: flag very few. Low sensitivity, high specificity. Moving along the curve from blue to red = increasing the threshold.

AUC — Area Under the Curve

AUC summarises the entire curve as one number — literally the area under it.

AUC	
1.0	Perfect — curve hugs the top-left corner
0.75	Good — meaningful discriminative ability
0.5	Random — curve is the diagonal
< 0.5	Worse than random — something is flipped

Picking the Best Threshold

There is no single "best" threshold — it depends on clinical priorities. A common default is **Youden's J**: maximise sensitivity + specificity simultaneously.

$$J = \text{sensitivity} + \text{specificity} - 1 \quad (\text{maximise this})$$

For OUD relapse prediction, missing a relapser (false negative) is probably more costly than a false alarm. This argues for prioritising **sensitivity** — accepting a lower threshold even if it produces more false positives.

In R (tidymodels)

Find the threshold that maximises Youden's J from cross-validated predictions:

```
rs_results |>
  collect_predictions() |>
```

```
roc_curve(truth = outcome, .pred_0, event_level = 'first') |>  
mutate(j = sensitivity + specificity - 1) |>  
slice_max(j, n = 1)
```

Output: .threshold (the cutoff probability), sensitivity, specificity, j at that point.